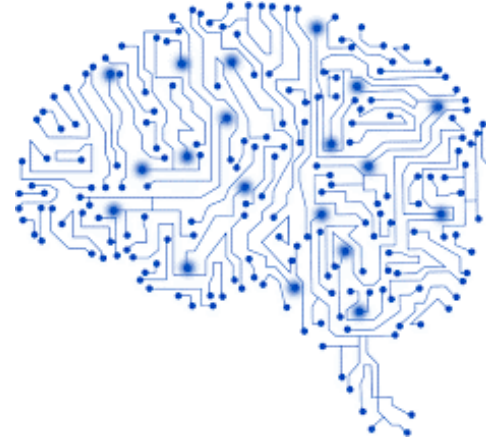




University of Minho
School of Engineering



Dados e Aprendizagem Automática

Introduction to Data Science Concepts

DAA 2026/2027

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Part I

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- Data Types
- Mean, Median & Mode
- Standard Deviation & Variance
- Probability Density Functions
- Percentiles
- Covariance & Correlation
- Virtual Environment
- Environment Setup
- Hands On



Data Types

Data Types

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- The most common data types:
 - Numerical
 - Categorical
 - Ordinal

Data Types

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Numerical

- It represents some sort of quantitative measurement
 - E.g., heights of people, page load times, and stock prices
- Discrete Data
 - It is integer-based; often represents the number of occurrences of an event
 - How many purchases did a customer make in a year?
 - How many times did I flip “heads”?
- Continuous Data
 - It can have an infinite number of values
 - How long did it take for a user to check out?
 - How much rain fell on a given day?

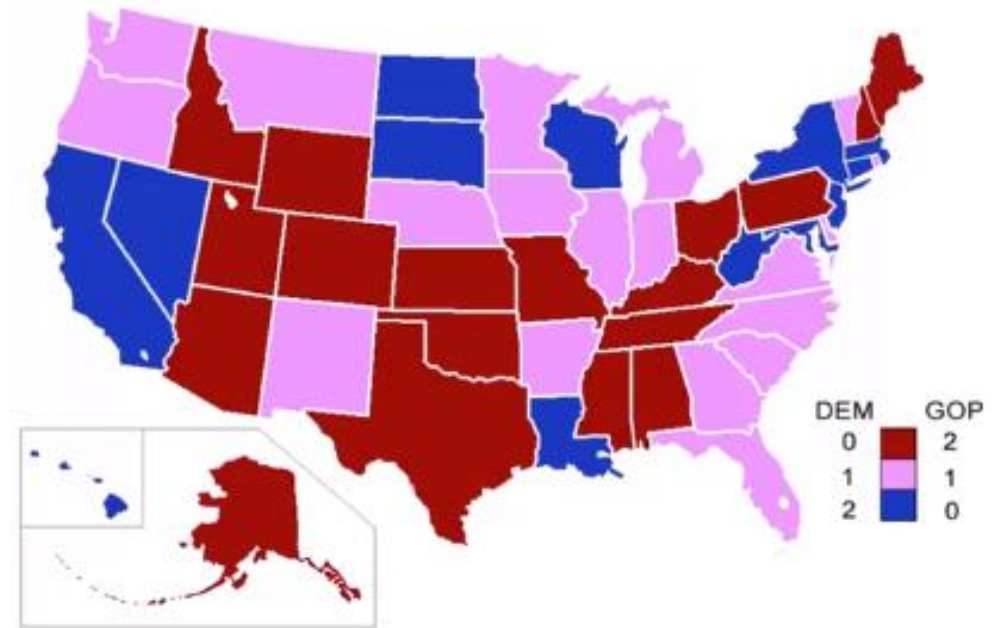


Data Types

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Categorical

- Qualitative data that has no inherent mathematical meaning
 - E.g., gender, yes/no (binary data), race, state of residence, product category, and political party
- Numbers can be assigned to these categories to represent them more compactly, but the numbers do not have any mathematical meaning



Data Types

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Ordinal

- A mixture of numerical and categorical data
- Categorical data with mathematical significance
- E.g., movie ratings on a scale of 1 - 5
 - Ratings must be 1,2,3,4 or 5
 - These values have a mathematical meaning: a rating of 1 means that the movie is worse than a movie with a rating of 2



Data Types

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Quick Quiz

- Classify the following types of data as numerical, categorical or ordinal.
 - How much gas/petrol is in your tank?
 - Rate your overall health on a scale of 1 – 4, where 1 is “poor”, 2 is “moderate”, 3 is “good” and 4 is “excellent”
 - The nationalities of your classmates
 - Ages in years
 - Money spent in a store





Mean, Median & Mode

Mean, Median & Mode

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Mean

- aka *average*
- Sum/number of samples
- Example:
 - The number of children in each house on a given street:

0, 2, 3, 2, 1, 0, 0, 2, 0

The MEAN is $(0+2+3+2+1+0+0+2+0)/9=$ **1.11**

Mean, Median & Mode

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Median

- Sort the values and select the one at the midpoint
- Example:

0, 2, 3, 2, 1, 0, 0, 2, 0

Sorting it:

0, 0, 0, 0, 1, 2, 2, 2, 3



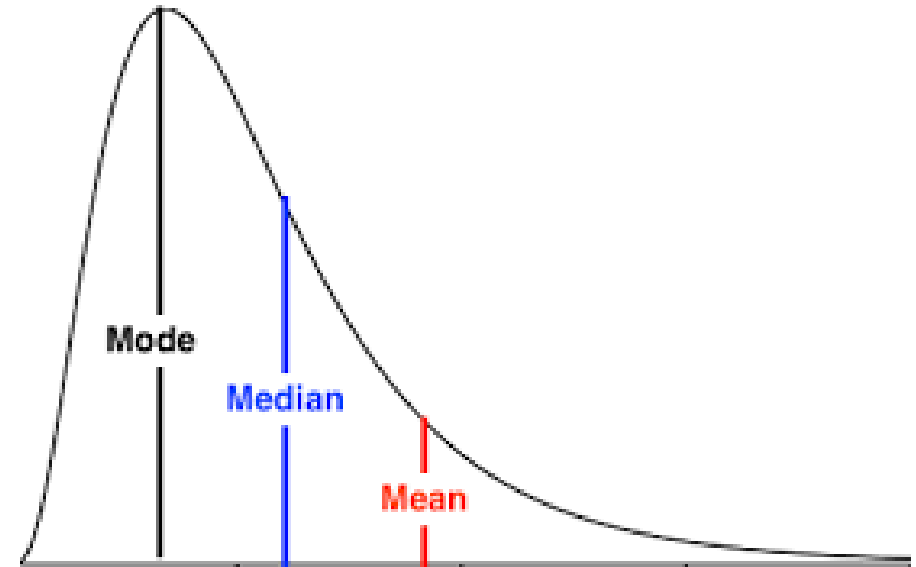
- If you have an even number of samples, take the average of the two in the middle.

Mean, Median & Mode

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Median

- The median is less susceptible to outliers than the mean
 - For example, the mean household income in the USA is \$72,641, but the median is only \$51,939 – because the mean is skewed by a handful of billionaires
 - In this example, the median better represents the “typical” American



Mean, Median & Mode

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Mode

- The most common value in a dataset
 - Not relevant to continuous numerical data
- The example of the number of children in each house:

0, 2, 3, 2, 1, 0, 0, 2, 0

How many of each value are there?

0: 4, 1: 1, 2: 3, 3: 1

The MODE is **0**

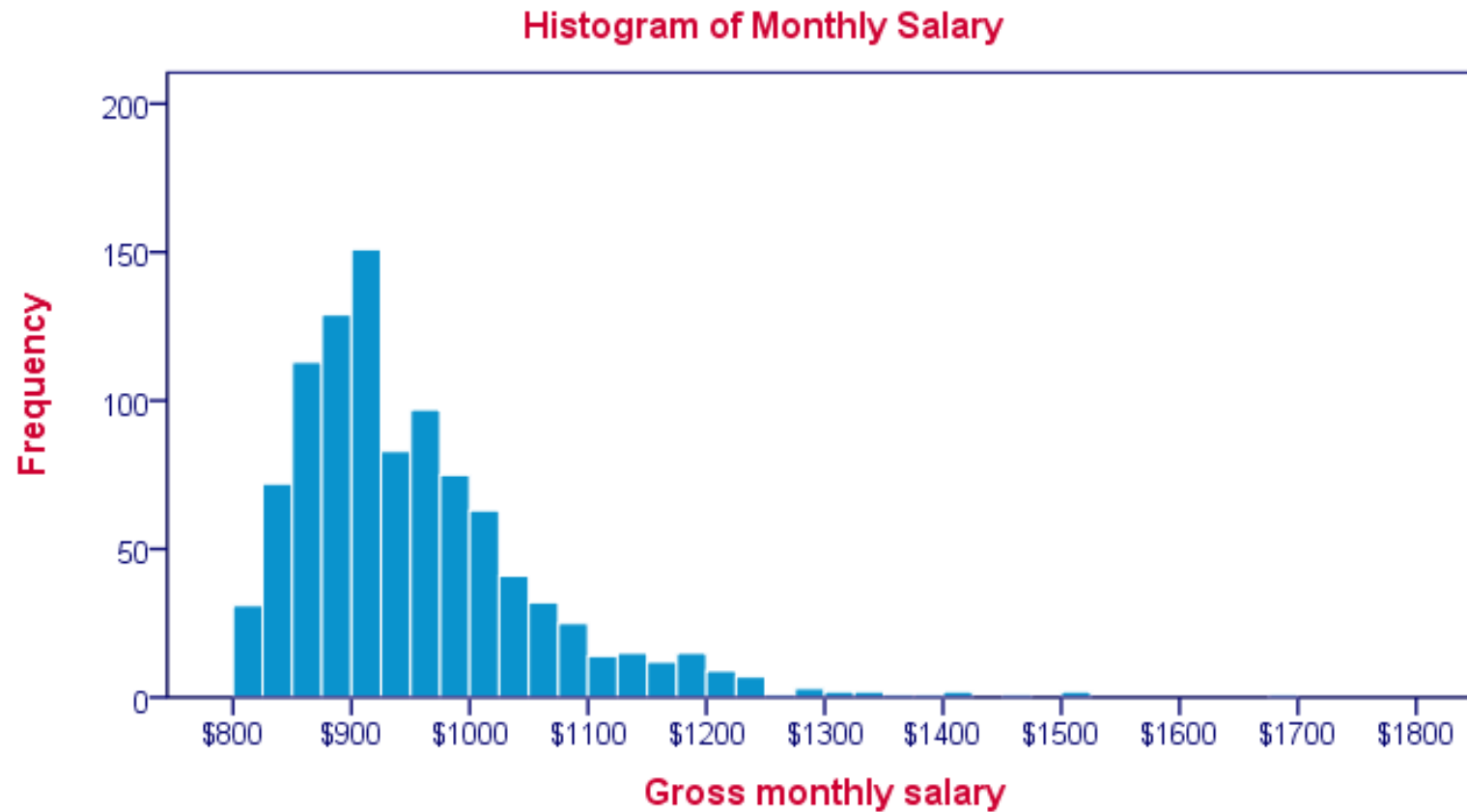


Standard Deviation & Variance

Standard Deviation & Variance

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Example of a histogram



Standard Deviation & Variance

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Variance measures how “spread-out” the data is

- Variance (δ^2) is simply the average of the squared differences from the mean
- Example:

What is the variance of the dataset **(1, 4, 5, 4, 8)**?

1. Find the mean: **(1+4+5+4+8) / 5 = 4.4**
2. Find the difference from the mean: **(-3.4, -0.4, 0.6, -0.4, 3.6)**
3. Find the squared differences: **(11.56, 0.16, 0.36, 0.16, 12.96)**
4. Find the average of the squared differences:

$$\delta^2 = (11.56+0.16+0.36+0.16+12.96) / 5 = \mathbf{5.04}$$

Standard Deviation & Variance

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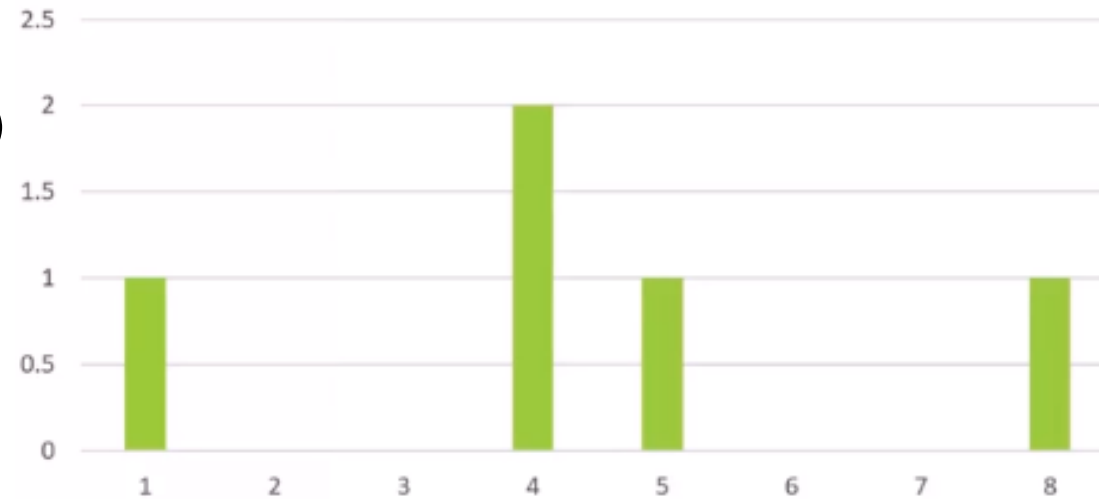
Standard Deviation, δ , is the square root of the variance

- It is usually used to identify outliers
- Data points lying more than one standard deviation away from the mean can be considered unusual
- The distance of a data point from the mean can be expressed in terms of “how many sigmas” it is away
- For example, given the case study = **(1,4,5,4,8)**

Mean = 4.4

$\delta^2 = 5.04$

$\delta = 2.24$





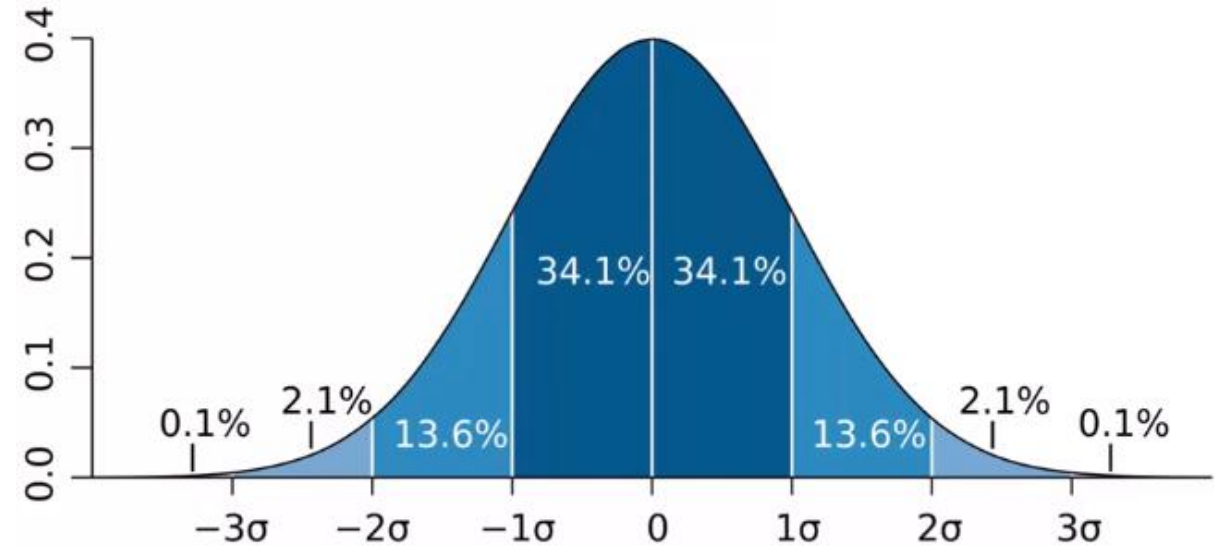
Probability Density Functions

Probability Density Functions

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“Normal Distribution”

- It gives the probability of a data point falling within a given range of a specific value
- A normal probability density function can be calculated based on histogram values

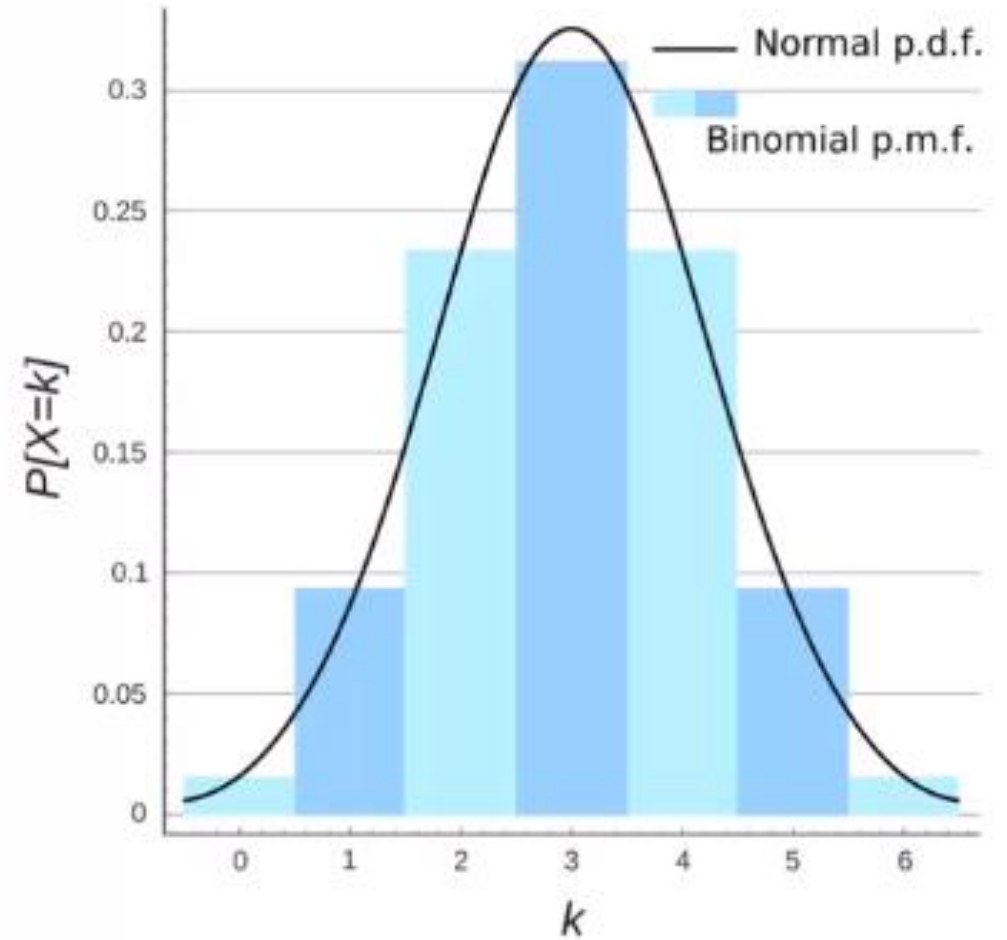


Probability Density Functions

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Probability Mass Function

- It is used for discrete data



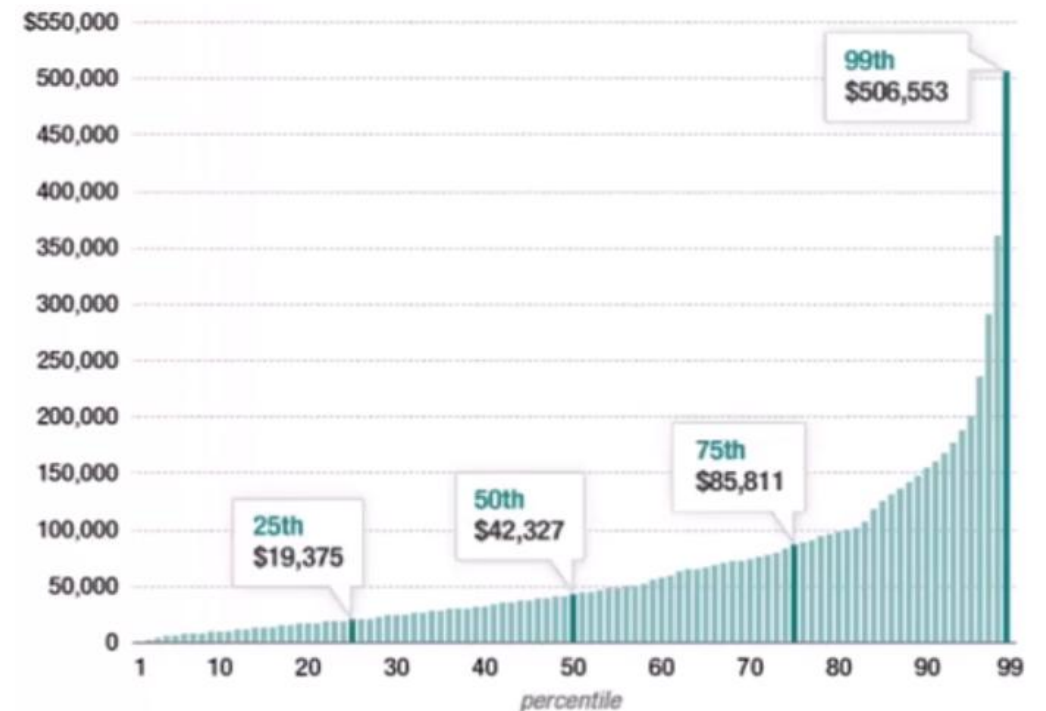


Percentiles

Percentiles

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- In a dataset, at what point are X% of the values less than that value?
- E.g., income distribution
 - Take all incomes from a country's population and sort them
 - The 99th percentile represents the income amount at which 99% of the population earns less than that value (i.e., \$506,553)

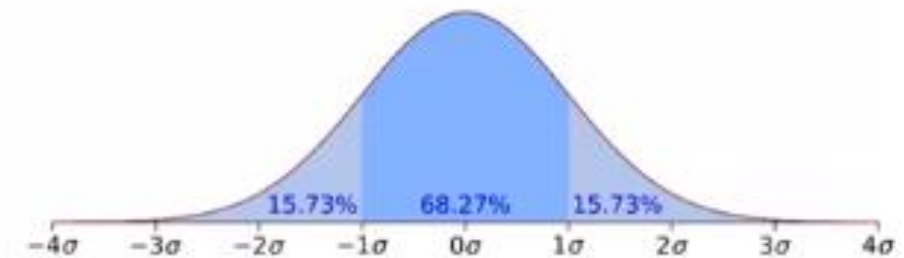
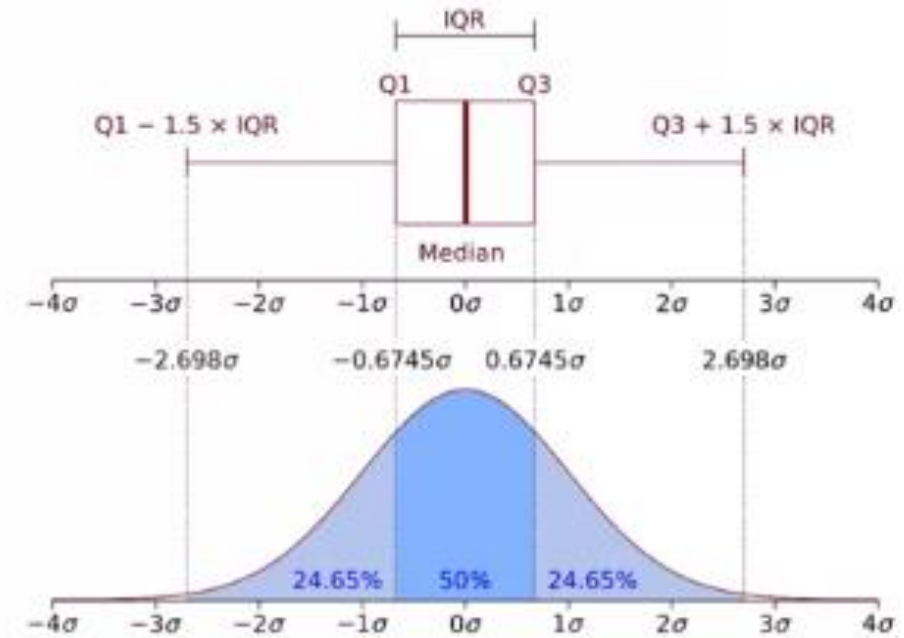


Percentiles

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Percentiles in a normal distribution

- The area between Quartile 1 and Quartile 3 represents 50% of the data distribution
- **IQR (Inter-Quartile Range)** represents the area in the middle of the distribution where the data is more concentrated





Covariance & Correlation

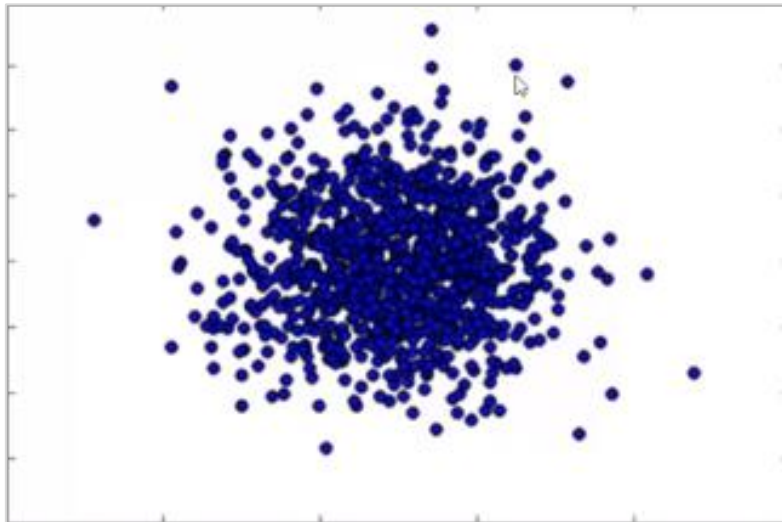
Covariance & Correlation

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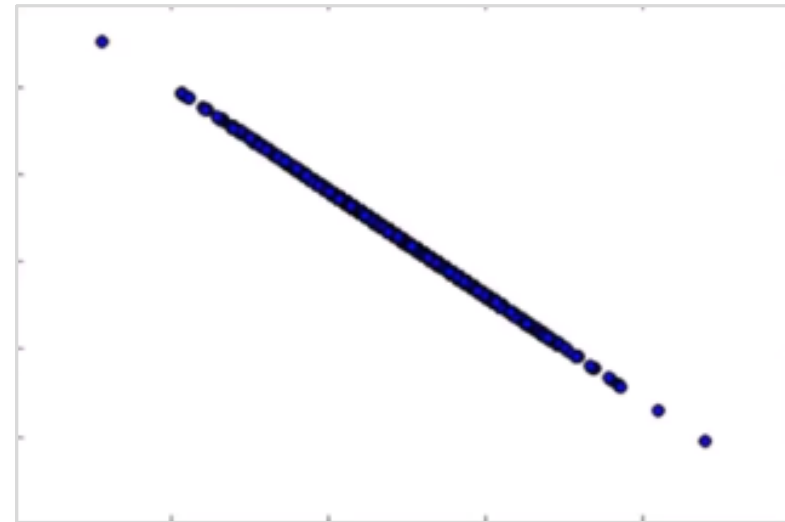
Covariance

- It measures how two variables vary in tandem from their means
- In other words, it shows how dependent are two attributes are on each other

Low Covariance



High Covariance



Covariance & Correlation

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Measuring it:

- Imagine the datasets for the two variables as high-dimensional vectors
- Convert these to vectors of variances from the mean
- Then, take the dot product (i.e., the cosine of the angle between them) of the two vectors
- Divide by the population size

Population Covariance Formula

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

Covariance & Correlation

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Interpreting **covariance** can be difficult

- A small covariance (close to 0) indicates a low degree of correlation between the two variables
- A large covariance (far from 0, and potentially negative for inverse relationships) indicates a correlation

Interpreting **correlation** is easier

- The normalization value is the covariance divided by the standard deviations of both variables
 - A correlation of -1 indicates perfect inverse correlation
 - A correlation of 0 indicates no correlation
 - A correlation of 1 indicates a perfect positive correlation

Covariance & Correlation

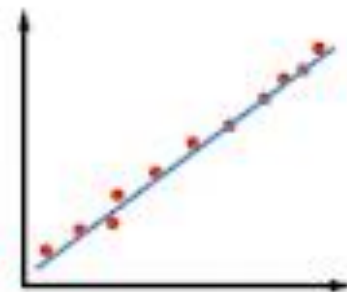
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Correlation does not imply causation!

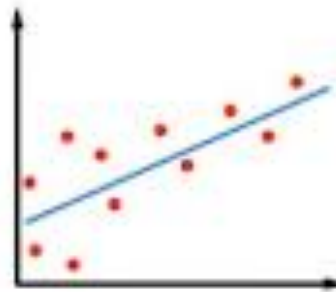
- Only a controlled, randomized experiment can provide insights into causation
- Use correlation to decide what experiments to conduct

Covariance & Correlation

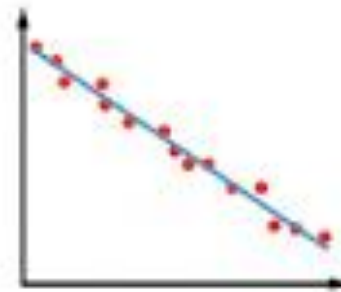
29



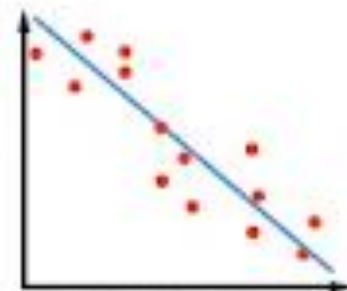
**STRONG POSITIVE
CORRELATION**



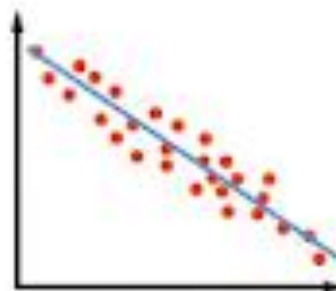
**WEAK POSITIVE
CORRELATION**



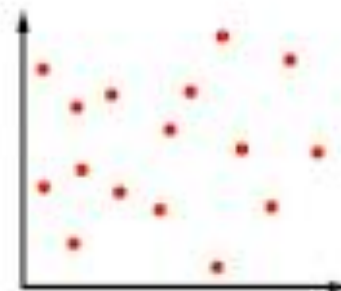
**STRONG NEGATIVE
CORRELATION**



**WEAK NEGATIVE
CORRELATION**



**MODERATE NEGATIVE
CORRELATION**



NO CORRELATION



Virtual Environments

Virtual Environments

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- Virtual environments allow to set up virtual installations of Python and its libraries on a local computer
- This means you can have multiple versions of Python and libraries, and easily activate or deactivate these environments
- Let's look at some examples of why you might want to do this

Virtual Environments

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- If you want to program in different versions of a library
- For example:
 - You develop a program with SciKit-Learn 0.17
 - Then, SciKit-Learn 0.18 is released
 - You want to explore the new version, but you don't want your old code to break
- If you may also want to ensure that your library installations are in the correct location
- For example:
 - You have multiple versions of Python on your computer
 - You might one environment with Python 2.7 and another with Python 3.6

- Conda has a built-in virtual environment manager that simplifies the entire process
- Since we don't require all of Conda's features, we suggest using **Miniconda** instead
- Check out the resource link for the official documentation:

<https://docs.conda.io/projects/miniconda/en/latest/>

- **Miniconda** is a free minimal Conda installer. It is a small bootstrap version of Anaconda that includes only Conda and Python, along with the packages they both depend on, and a few other useful packages (such as Pip and Zlib)
- If you require additional packages, you can use the `conda install` command to install from the thousands of packages available by default in Anaconda's public repository

Virtual Environments

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- Command prompt usage example:

List all packages and versions installed	<code>conda list</code>	
List environments	<code>conda info --envs</code>	
Create <i>myenv</i> environment	<code>conda create --name myenv python=3.14 numpy</code>	
Activate <i>myenv</i> environment	<code>conda activate myenv</code>	
Enter Python	<code>python</code>	
Import libraries	<code>import numpy as np</code>	
	<code>import pandas as pd</code>	-> Error
Exit Python	<code>quit()</code>	
Installation of Pandas library	<code>conda install pandas</code>	
Deactivate environment	<code>conda deactivate</code>	
Delete <i>myenv</i> environment	<code>conda env remove --name myenv</code>	

Virtual Environments

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- **Python** also has a built-in virtual environment manager whose main advantage is that it simplifies library management across several projects
- However, it only includes the default Python packages, so all library installations are left up to the user

Environment creation	<code>python -m venv <venv_name></code>
Activation	macOS/Linux <code>. ./<venv_name>/bin/activate</code>
	Windows (PS) <code>.\<venv_name>\Scripts\Activate.ps1</code>
	Windows (cmd) <code>.\<venv_name>\Scripts\activate.bat</code>
Package installation	<code>pip install <package_name></code>
Environment deactivation	<code>deactivate</code>
Get package information	<code>pip freeze > requirements.txt</code>



Virtual Environments

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- **Windows Subsystem for Linux (WSL)** allows you to run a Linux environment in a Windows machine, enabling its features (Linux command-line tools, utilities and applications) without the need for a virtual machine or dual-booting
- How to install it:
 1. Open Windows PowerShell as an administrator
 2. Enter the following command: `wsl --install`
 3. Restart the computer
- Installation guides:
 - <https://learn.microsoft.com/en-us/windows/wsl/install>
 - <https://ubuntu.com/wsl>
- For older versions of WSL, see <https://learn.microsoft.com/en-us/windows/wsl/install-manual>



Environment Setup

Environment Setup

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- In this course we will use Jupyter Notebooks/Spyder for teaching purposes and to provide notes
- Please **note** that you are free to use **whichever development environment you prefer** (e.g., Spyder, PyCharm or VisualStudio)
- We will be using the Python 3.14 through the Miniconda distribution for this course
See <https://www.python.org/downloads/>

Environment Setup

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- If you are an experienced user who already has Python:
 - You may wish to install Jupyter and the required APIs using the Python package manager pip instead of Miniconda
 - Simply open your command prompt or terminal and enter the following command:

```
pip install jupyter
```

- We highly recommend that new users install Miniconda or Anaconda:
 - This will conveniently install Python, the Jupyter Notebook and other commonly used packages for scientific computing and data science
 - Visit www.jupyter.org to walk through the installation steps!

Environment Setup

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The screenshot shows the Jupyter.org homepage in a web browser. The browser's address bar displays "https://jupyter.org". The page features a navigation bar with links: Try, Install, Community, Subprojects, AI, Documentation, News, Social, Governance, Security, and About. The main content area has a large Jupyter logo in the center, surrounded by various programming language and framework icons including JS, Python, R, Julia, C#, F#, Spark, and others. Below the logo, the text reads: "Free software, open standards, and web services for interactive computing across all programming languages".

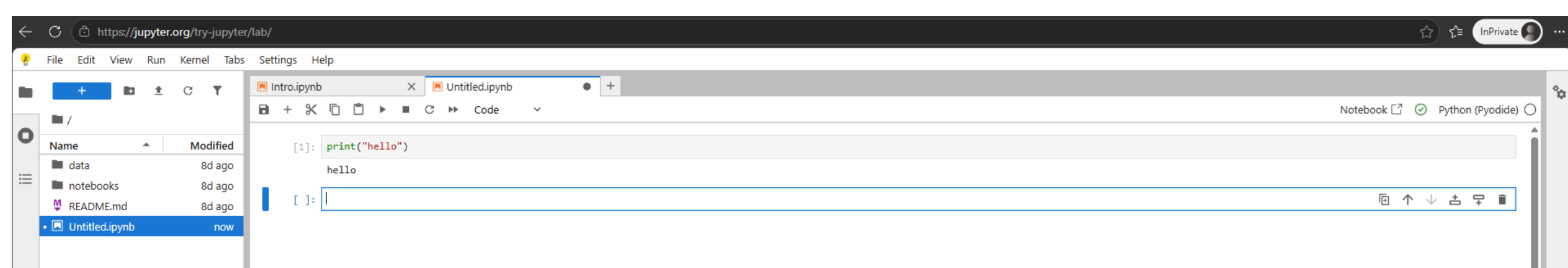


JupyterLab: A Next-Generation Notebook Interface

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible

Environment Setup

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Hands On

Hands On

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T1

- We will use Scikit-learn/Sklearn
- Download and install the Miniconda package for your operating system (Windows, macOS, Linux) from <https://www.anaconda.com/docs/getting-started/miniconda/system-requirements>
 - Miniconda – Python 3.14.7
 - Deep Learning libraries that are **not** required: Theano, Tensorflow, Keras
 - You will need to install Python (<https://www.python.org/downloads/>)
 - Setup guides (for reference):
 - <https://www.guvi.in/blog/how-to-setup-a-python-environment-for-machine-learning/>
 - <https://www.machinelearningplus.com/machine-learning/setup-python/>
 - <https://machinelearningmastery.com/setup-python-environment-machine-learning-deep-learning-anaconda/>

Hands On

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T2

- Start the Miniconda prompt and create a Python3.14 environment:

- Open the terminal and execute:

```
conda create --name envNAME python==3.14
```

- To work inside the Python environment, execute:

```
conda activate envNAME
```

- To install packages, enter the environment and execute:

```
conda install PACKAGENAME
```

- To exit the Python environment, execute:

```
conda deactivate
```



```
numpy  
pandas  
matplotlib  
seaborn  
scikit-learn  
jupyterlab
```

Hands On

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T3

- Activate the created virtual environment you created and check that the libraries have been installed
- Validate the installation of the library set presented in **T2**

T4

- Briefly check the documentation for each library mentioned in question **T2**
- Identify their relevance in the context of Machine Learning algorithm development